MCTR 426 3D Printing and Designing

Max. Marks : 100	L	T	P	C
(CA: 40 + ESA: 60)	4	0	0	4

Learning Outcomes

The students will be able to:

- Understand the basic concepts of 3D printing
- Understand the difference in various 3D printing techniques
- Model a 3D design and analyze the structure
- Acquire the optimization skills for designing and fabrication
- Analyze a fabricated design to optimize a design model
- Learn about the maintenance and troubleshooting of a printing system.

Section A

Introduction 3-Dimensional printing (3DP): 3D Printing vs traditional techniques, Advantages of 3D Printing, Basic Introduction to solid modeling, designing, Introduction to STL file, slicing and layering.

Costs of 3D printers, Cost estimation of 3DP, Introduction to industrial, educational and research, household and personal 3D Printers, Introduction of open-source printers, building and developing 3D printer for custom applications.

3D printing technologies: Fused filament fabrication (FFF)- Fused Deposition Modeling (FDM), Stereolithography (SLA), Selective Laser Melting (SLM), Binder Jetting, Selective Laser Sintering (SLS), Inkjet Printing: Diamatix and household, Aerosol jet printing, Polyjet printing, Digital Light Processing (DLP)

Materials and substrates: Form of materials, Polymers, metals, ceramic, others

Section B

Modeling in 3-Dimensions: Software basic principles, boundary conditions, machine limitations

Modeling software and their features: AutoCAD, 123D, introduction to other freely available software, Point cloud to STL Conversion.

Pre-processing: Data Preparation, STL error removal, Printing technique identification, Printing parameter selection, Build orientation, Material selection, Printing system limitations, Cost-analysis.

CAD model optimization: Introduction to design optimization, requirements of optimization, CAD model preparation for 3DP, Parametric 3D modeling, CAD model modification for printer limitation.

Model infill: Requirement, Necessity, types, realization. Printer calibration

Slicing of CAD model: Type of slicing, Slicing algorithm

Analysis of a failed model for design corrections, Troubleshooting a printer system Introduction to Post-processing requirements

Maintaining a 3D printer: System maintenance, extruder maintenance, motor maintenance, new and old material maintenance, pressure system maintenance.

Section C

Advanced parametric 3D Modeling: Modeling complex shapes and structures, software and system tolerances.

Customizing an advanced design print: build plate adhesion, infill adjustments, print speed modifications, corner speed alterations, support structures.

Post-processing requirements: Cleaning, support structure removal, sanding, adhesion, mounting, curing, heating, smoothing, polishing and electroplating.

Industrial printer systems: Polymer-based printers, metal-based printers, a brief introduction to other printers.

Multi-extrusion systems: printing with multiple polymers, printing with multiple materials, printing with metals.

Latest advancements in industrial 3D printing.

Applications and case studies: Engineering applications, Medical applications, Reverse Engineering.

Text Books:

1. Chua, C.K., Leong, K.F., Rapid Prototyping: Principles and Applications in Manufacturing, John Wiley and Sons Inc., 2000.

Reference Books:

1. Pham, D.T., Demov, S.S., Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling, Springer-Verlag London Limited, 2001.
2. Hopkinson, N., Hague, R.J.M. and Dickens, P.M., Rapid Manufacturing and Industrial Revolution for the Digital Age, John Wiley and Sons Ltd, Chichester, 2005.
3. Noorani, R., Rapid Prototyping: Principles and Applications, John Wiley & Sons, Inc., New Jersey, 2006.
4. Zeid, I., Mastering CAD/CAM, Tata McGraw Hill, 2006
5. Gebhardt, A., Rapid Prototyping, Hanser Gardner Publications, Inc., Cincinnati, 2003.
6. Gibson, I., Software Solutions for Rapid Prototyping, Professional Engineering Publication Ltd., 2002.
7. Patri, K. V., and Weiyin, Ma, Rapid Prototyping - Laser-based and Other Technologies, Kluwer Academic Publishers, U.S.A., 2003.

e-Resources:

1. https://nptel.ac.in/courses/112/103/112103306/
2. https://nptel.ac.in/courses/112/104/112104265/

IT 412 Internet of Things

Max. Marks : 100	**L**	**T**	**P**	**C**
(CA: 40 + ESA: 60)	**4**	**0**	**0**	**4**

Learning Outcomes:

After successful completion of the course students will be able to

- Understand the concept of IoT.
- Understand what constitutes an IoT design solution.
- Identify the sensors and basic electronic design needed for different IoT solutions.
- Analyze basic proptocols of IoT.
- Implement basic IoT applications on Arduino and Raspberry Pi to provide IoT solutions for various domains.

Section A

Introduction of IoT, Sensing, Actuation, Sensor Networks, Machine-to-Machine Communications M2M to IoT –Introduction, industrial structure for IoT, architecture for conversion of M2M to IoT, design principles ,capabilities of IoT, IoT network architecture , standard protocols, IoT Architecture Reference Model, Introduction to SDN, SDN for IoT, Data Handling and Analytics, Introduction to Cloud and Fog Computing, Sensor-Cloud. Domain specific applications of IoT: Home automation, Industrial applications, Surveillance applications, other IoT applications.

Section B

IoT Access Technologies: Physical and MAC layers, Bluetooth, RFID, WiMax, topology and Security of IEEE 802.15.4, 802.15.4g, 802.15.4e, 1901.2a, 802.11ah and LoRaWAN – Network Layer: IP versions, Constrained Nodes and Constrained Networks – Optimizing IP for IoT: From 6LoWPAN to 6Lo, Routing protocols (Routing over Low Power and Lossy Networks) Application Transport Methods: Supervisory Control and Data Acquisition, Application Layer Protocols: CoAP and MQTT, Security in IoT protocols.

Section C

Introduction to concept of IoT devices, IoT configurations, basic components, networking, sensors, introduction to Edge computing and Embedded IoT. Introduction to Arduino, types of Arduino, Arduino toolchain, Arduino programming structure, Sketches, Pins, Input-output from pins using sketches, Introduction to Arduino shields, Introduction to Raspberry-Pi microcomputer Accessing GPIO pins, Sending and receiving signals using GPIO pins

Suggested Books:

1. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014.
2. Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-on-Approach)", 1stEdition, VPT, 2014.
3. Francis daCosta, "Rethinking the Internet of Things: A Scalable Approach to Connecting Everything", 1st Edition, Apress Publications, 2013

Suggested E-Learning Material:

1. https://onlinecourses.nptel.ac.in/noc19_cs31
2. https://www.edx.org/course/introduction-to-the-internet-of-things-iot-1

RS 401 Geoinformatics

Max. Marks : 100	**L**	**T**	**P**	**C**
(CA: 40 + ESA: 60)	**4**	**0**	**0**	**4**

Learning Outcomes:

After successful completion of the course students will be able to

- Describe spatial database, Co-ordinate and projection system
- Analyze vector and raster based analysis in Geographical Information Sciences
- Describe different types of satellite system and digital image processing
- Describe global cover based global position systems i.e. GPS, GLONASS
- Describe applications of remote sensing and GIS in natural resources management

Section A

Geographical Information System: Definition, Components of GIS, Hardware and software requirements for GIS, Coordinate system and projections, Database structure and formats, Spatial data models - raster and vector. Data inputting, Data base design - editing and topology creation. Linkage between spatial and Non spatial data, Query (Attribute/Spatial), Vector based analysis. Raster based analysis. Errors, Digital Elevation Model, Network analysis, O'pen source and WebGIS.

Section B

Remote Sensing: Definition - components of remote sensing - energy sensor, interacting body; Type - active and passive remote sensing. Satellite System - meteorological, communication and remote sensing. Platforms - aerial and space, synoptivity and repeativity. Electromagnetic Radiation (EMR) - EMR spectrum- visible, infrared [IR) middle IR, thermal IR and microwave. EMR interaction with earth surface material, radiance, irradiance, incident, reflected, absorbed and transmitted energy, spectral response pattern - spectral signature curves (water, soil and vegetation].

Digital Image Processing : Digital Image, Satellite Image - characteristics and formats. Resolution - spatial, spectral, radiometric and temporal; Introduction to rectification, enhancement; Classification - Unsupervised and Supervised classification.

Section C

Global Positioning System: Global Navigation Satellite System (GNSS), GPS, GLONASS, GALILEO, Segments - space, control, user, GPS Satellite signals, sources of errors and corrections.

Applications of Remote Sensing and GIS:

Applications of GIS and Remote Sensing in resource management (forestry, agriculture, urban telecommunication, transportation, water resources and environment).

Suggested Books:

1. Lo, C. P., & Yeung, A. K. (2002). *Concepts and techniques of geographic information systems*. Upper Saddle River, NJ: Prentice Hall.
2. Ian, H. (2010). *An introduction to geographical information systems*. Pearson Education India.
3. Joseph, G. (2005). *Fundamentals of remote sensing*. Universities press.

4. Jensen, J. R., & Lulla, K. (1987). *Introductory digital image processing: a remote sensing perspective*. Prentice Hall.
5. Sabins, F. F., & Sensing, R. (1987). *Principles and interpretation: Remote sensing.* San Francisco.

Suggested E-Learning Material:

1. GIS in Civil Engineering
   https://nptel.ac.in/courses/105102015/
2. Geospatial Information and Services
   http://www.oc.nps.edu/oc2902w/gis/gisdemo/

CS 508 Big Data Analytics
Max. Marks : 100
(CA: 40 + ESA: 60)

Learning Outcomes:
After successful completion of the course students will be able to

- Understand big data analytics and identify the main sources of large scale big data in the real world.
- Learn various frameworks like Hadoop. NOSQL to efficiently store retrieve and process Big Data for Analytics.
- Implement several Data Intensive tasks using the Map Reduce Paradigm in Hadoop.
- Program applications using tools like Hive, pig, NO SQL for Big data Applications.
- Construct scalable algorithms for large Datasets using Map Reduce techniques.
- Apply the knowledge of Big Data gained to fully develop DBA applications for real life applications.

Section A

Introduction to Big Data Analytics, Evolution, Structuring Big Data, Types of Big Data, 4Vs, Big Data Analytics: Advantages, Applications. Comparing Report and Analysis. The Analytic Process, Types of Analytics. Characteristic of Big Data Analytics. Framing the Problem for Analytics. Statistical Significance or Business Importance of Analytics. Making Inferences. Analytic Approaches: History and Ensemble Methods, Graphical User Interface, Data Visualization for Big Data.

Section B

Gathering Data on a Distributed Environment. Architecture, Features of Hadoop Framework: HDFS, Map Reduce, YARN, Hbase, Hive, Sqoop,Zookeepker, Oozie. Exporting Data to HDFS and Importing Data from HDFS, HDFS Commands. HBase Architecture, Storing Big Data with Hbase, Interacting with Hadoop Ecosystem, Combining HBase and HDFS.
MapReduce Framework, Working of MapReduce, Techniques to Optimize MapReduce Jobs. Building and Executing Applications. Controlling MapReduce Execution with InputFormat, Taking Input from files and applying operations for customization.

Section C

Understanding Hive, Hive Variables, Properties, Queries and Data Types. Built in Functions in Hive. Working with Databases in Hive: Creating, Viewing, Droping and Altering. Creating and

Modifying Tables. Using Hive DDL Statements and DML Statements. Executing HiveQL. Applying Joins, Group By and Order By clauses.
Pig Architecture. Properties of Pig, Running Pig Programs, Working with Operators in Pig, Working with Functions in Pig.

Suggested Books:

1. White, Tom. (2012). *Hadoop: The definitive guide*. O'Reilly Media, Inc.
2. Miner, D. & Shook, A. (2012). *MapReduce Design Patterns: Building Effective Algorithms and Analytics for Hadoop and Other Systems*. O'Reilly Media, Inc.
3. Loshin, D. (2013). *Big data analytics: from strategic planning to enterprise integration with tools, techniques, NoSQL, and graph*. Elsevier.

Suggested E-Learning Material:

1. Big Data & Analytics by Kent State University
   https://www.cs.kent.edu/~jin/BigData/index.html
2. Big Data Specialization by University of California San Diego
   http://www.coursera.org/specializations/big-data

AI 504 Cognitive Computing

Max. Marks: 100	L	T	P	C
(CA: 40 + ESA: 60)	4	0	0	4

Learning Outcomes:

After successful completion of the course students will be able to

- Understand Computational Theories of Human Cognition.
- Apply Inductive Principles Required for Knowledge Acquisition
- Analyze and Design Real World Problems to Understand and Implement Dynamic Behaviour of a System

Section-A

Elements of Cognitive System. Understanding the Complex Relationships Between Systems. Understanding Cognition, Transformation of Artificial Intelligence into Cognitive Computing System. Uses of Cognitive Computing Systems, Gaining Insights from Data, Bringing Data into Cognitive System, Defining Objectives, Defining Domain, Understanding the users and defining their attributes, defining questions and exploring insights, training and testing of cognitive computing systems.

Elements of Cognitive Models, Models of Working Memory: Decay Process, Recall Process, Rehearsal, Sufficiency of a Cognitive Model, Verisimilitude Versus Truth, Nature of Explanations.

Section-B

Perception and Attention: Principles of perception, Different theoretical approaches to perception (Gibson, Helmholtz, Gestalt, etc), Color Perception, Perceptual organization, Object recognition, Motion and Time perception, Selective Attention, Sustained Attention, Divided Attention,Executive Control.

Learning and Memory: Principles of classical conditioning and operant conditioning, Theories of Learning, Reinforcement schedules, Skill Acquisition and Performance, Sensory memory, Working Memory, Models of Semantic Memory, Autobiographical Memory, Retrieval, Forgetting, Implicit learning and memory.

Section-C

Reasoning Cognitive Systems through Probabilistic Learning Bayesian Networks, Approximate Inference, Constructing Bayesian Networks,

Markov Chains, Hidden Markov Model: BaumWelch algorithm, Viterbi Algorithm.

Language processing in Cognitive Computing Systems: The Phonological Loop, Importance of Context, Understanding Linguistics, Techniques For Resolving Ambiguities at Different Levels.Applications of Cognitive Computing, Use Cases of Cognitive Computing Systems: Enhancing the shopping experience, Connected World of Internet of Things, Understanding your customer, fraud detection. IBM Watson Framework.

Suggested Books:

1. Russell, S. J., &Norvig, P. (2016). Artificial intelligence: a modern approach. Pearson.
2. Vermon D. (2014). Artificial Cognitive Systems: A Premier, MIT Press.
3. Wang, Y., Zhang, D., &Kinsner, W. (Eds.). (2010). Advances in cognitive informatics and cognitive computing (Vol. 323). Springer.
4. Nilsson, N. J. (1998). Artificial intelligence: a new synthesis. Morgan Kaufmann